Contents

1 day 11: data and text analysis **1**
1.1 Introduction to pandas 1
1.2 other libraries for plotting 2
1.3 Intermediate pandas (if time) 3
1.4 text analysis with NLTK 4

1 day 11: data and text analysis

Agenda

- data analysis with pandas
- BREAK
- text analysis with NLTK

1.1 Introduction to pandas

- df = pd.read_csv('https://bit.ly/664-met-woman')
- viewing data
  - info(), head(), tail(), shape, columns
  - selecting columns
    - syntax: either dot or brackets
    - column = Series type of data
  - values
    - the items in a column
    - gives back a np object
  - slicing rows, df[100:110]
  - describe() # statistics
- sorting data
  - value_counts('title') # department, artist display name
    - counts most common values for a column
  - sort_values('accessionYear')
    - sorts by values in a column

- filtering data
  - first, create a boolean
    - df['title'] == 'Woman'
      - this is a conditional
  - then, create a variable to hold the booleans
    - woman = df['title'] == 'Woman'
      - this is a variable assignment from a conditional
  - finally, creates new dataframe, df[woman]
    - we can now further sort elements from this dataframe.
      - df[woman]['artistDisplayName']
      - df[woman].value_counts('department')
  - highlights = df['isHighlight'] == 1.0
    - by boolean condition
  - df[highlights].info()
  - courbet = df['artistDisplayName'].str.contains('Courbet', na=False)
    - using str.contains
  - df[courbet]['title']
- plotting data
  - df.plot(kind='bar')
  - df.value_counts('department').plot(kind='bar')
  - adding nlargest(10)
    - df.value_counts('title').nlargest(10).plot(kind='bar')
  - look at the docs, and explore some of the parameters to make the graph more detailed
    - df.value_counts('artristDisplayName').nlargest(10).plot(kind='barh', xlabel='Artist Name', title='Works of "Women" at the MET')
    - df.value_counts('').plot(kind='pie')

1.2 other libraries for plotting

- seaborn
- plotly

1.3 Intermediate pandas (if time)

- df = pd.read_csv('https://bit.ly/anti-trans-tracker-csv')
- sorting values
  - df.sort_values('Bill Type', inplace=True)
  - see "pandas.DataFrame.sort_values" in docs to understand other parameters
  - sorted = df.sort_values(['Bill Type', 'State'])
    - can sort by multiple values, create new df.
- filtering values
  - looking for different kinds of bans.
  - df['Bill Type'] == 'Trans Bathroom Ban'
- CHALLENGE: Which states have the most book bans?
- two ways:
  - df['Bill Type'] == 'Book Ban'
    - books = df['Bill Type'] == 'Book Ban'
    - df[books]
    - df[books].value_counts('State')
  - `str.contains`: filtering by words
    - df['Bill Type'].str.contains('Bathroom')
    - bathrooms = df['Bill Type'].str.contains('Bathroom')
    - df[bathrooms].info()
    - df[bathrooms].value_counts('State')
- plotting data
  - df.plot(kind='bar')
  - df.value_counts('Bill Type').plot(kind='bar')
  - adding nlargest(10)
    - df.value_counts('Bill Type').nlargest(10).plot(kind='bar')
  - look at the docs, and explore some of the parameters to make the graph more detailed

    - df.value_counts('Bill Type').nlargest(10).plot(kind='barh', xlabel='Number of Bills', title='Most Frequent Categories for Anti-Trans Bills')
    - df.value_counts('Bill Type').plot(kind='pie')

BREAK

1.4 text analysis with NLTK

- install and import NLTK
- create sentence, then tokenize
    - why do we tokenize?